



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : MAC F211

Course Title : Linear Algebra and its Applications

Instructor-in-charge :JITENDER KUMAR
(email-kumar.jitender@pilani.bits-pilani.ac.in , chamber no.-6065-A)

1. Course description:

Quick review of Vector Spaces and Linear Transformations, Basis and Dimension, rank-nullity theorem, change of basis; Inner product spaces, Cauchy Schwarz's inequality, GramSchmidt orthonormalization, QR decomposition, Discrete Fourier Transform, and their applications, Spectral Theorem, Cayley-Hamilton Theorem, Decomposition theorem, and Jordan canonical forms, applications of SVD in Principal Component Analysis, use of SVD in low-rank approximations and to find least-squares solutions.

2. Scope and Objective:

Linear algebra has applications in pure and applied mathematics, computer science, economics, engineering, mathematical biology, operation research, physics, and statistics. Linear algebra and matrix methods are also important for data analysis. Linear Algebra and its applications typically aim to equip students with a solid understanding of fundamental concepts in linear algebra and the ability to apply these concepts to the above fields.

3. Course learning outcomes: After completion of this course students should be able to

CLO1.: Apply the Gram-Schmidt process to construct orthonormal bases in inner product spaces.

CLO2. Utilize Cauchy-Schwarz inequalities to prove geometric boundaries and inner product behaviors.

CLO3. Factor a matrix into orthogonal and upper triangular components for stable numerical computations.

CLO4. Implement the Spectral Theorem and structural decomposition theorems

CLO5. Formulate and compute optimal least-squares solutions for linear models using pseudoinverses derived from SVD.

CLO6. Apply Singular Value Decomposition (SVD) to execute Principal Component Analysis (PCA) on high-dimensional datasets.

4. Text books and references:

TB1. David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and Its Applications, 5th Edition, Pearson Education, 2023.

TB2. Gilbert Strang, Linear Algebra and Its Applications, Cengage India Private Limited; 4th edition, 2005.

RB1. Biswa Nath Datta. Numerical Linear Algebra and Applications, SIAM, 2010.

RB2. M.T. Nair and A.Singh, Linear Algebra, Springer Nature, Singapore Pte Ltd., 2018

RB3. K. Hoffman and R. Kunze, Linear Algebra, Prentice Hall Intl. Inc., Englewood Cliffs, 1971.

5. Course Format & Weekly Workflow

Wk no.	Module no.	Description	Ref.	Learning Outcome
1	1	L1-L4: Quick Review of Vector Spaces and Linear Transformations: Basis and Dimension, rank-nullity theorem, Change of basis	T1: Chapters 1,2,4	Overview of the basic required concepts of linear algebra
2	2	L5-L7: Inner product, Orthogonal vectors, Orthogonal sets	T1: Chapter 6 6.1, 6.2	Understand the concept of inner product and orthogonal sets
3-4	3	L8-L12: Orthonormal basis, Orthogonal projection, Gram-Schmidt orthonormalization, Inner product spaces: Cauchy	T1: Chapter 6 6.3, 6.4, 6.7	Understand the concept of Gram-Schmidt orthonormalization and Inner product

		Schwarz's inequality, Best approximation in inner products		spaces with properties
5	4	L13-L14: QR decomposition and Least-square problems	T1: Chapter 6 6.4, 6.5	Understand the concept of QR decomposition.
5-6	5	L15-L16: Applications of QR: Solution of $Ax=b$, Projections	R1: Chapter 7.6	Application of QR factorization
6	6	L17-L18: Applications of Least-squares problem, Discrete Fourier Transform	T1: Chapter 6, 6.6 T2: Sec 3.5	Understand the concept of Discrete Fourier Transform, and their applications.
7	7	L19-L21: Eigenvalues and Eigenvectors of real and complex matrices, Diagonalization of a Matrix	T2: Chapters 5.1, 5.2, 5.5	Understand the concept of diagonalization of a matrix
8-9	8	L22-L25: Similar transformations, Spectral Theorem, Cayley-Hamilton Theorem, Jordan form	T2: Chapter 5.6	Understand the Spectral Theorem, Cayley-Hamilton Theorem, Decomposition. theorem, and Jordan canonical forms.
9	9	L26-L27: Norms of vectors, matrices, Condition number	T2: Chapter 7 R1: Chapter 2.5	Understand the concept of vector and Matrix norms with properties
10	10	L28-L29: Singular Value decomposition and Singular vectors	R1: Chapter 7.8	Explore SVD and Singular vectors.
11-12	11	L30-L35: SVD Applications: Reduced SVD, norms, condition numbers, rank via SVD, Low rank approximation	R1: Chapter 7.8	Explore the use of reduced SVD in low-rank approximations.
13	12	L36-L38: Principal Component Analysis	T1: 7.5, Lecture notes	Understand the applications of SVD in Principal Component Analysis (PCA).

14	13	L39-L42: Pseudo-inverse, Least-squares solutions to linear systems	R1: Chapter 8 and lecture notes	Understand how SVD can be used to find least-squares solutions.
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6. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Midsemester Exam	30%	90	As per Timetable	Closed Book
Quizzes (03 Quizzes) *	30%	TBA	Quiz-1(M): 07.09.2026 Quiz-2(Th): 01.10.2026 Quiz-3(M): 16.11.2026	Closed Book
Comprehensive Exam	40%	180	As per Timetable	Closed/Open Book

: Score of **Best-2 Quizzes will be considered for grading.*

Criterion for NC: If a student gets less than 10% of the total marks, then NC will be awarded.

- 7. Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
- 8. Chamber consultation hours:** Monday, 5.00-5.50 PM (Office no. 6065-A)
- 9. Makeup Policy:** [Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.]

Instructor-in-charge;

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